Repository Tables and Figures: Calibrating Recall Across Operational Situations in Cross-Camera Target Search

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I. EARLIER EVALUATION ON THE HELD-OUT PERIOD

This appendix reports the evaluation that preceded the confirmation experiment. It used a pass/fail criterion with thresholds that were later abandoned (Section V-B); the results are reported as they were obtained. The criterion was replaced because the result rested on one evaluation seed and a narrow margin in the rarest group, the held-out period had been reused for further analyses (Section IV of the supplementary material), and the fixed thresholds were conventions. A reanalysis of the same period in the format of Section V-D is given at the end of this appendix.

A. PERIODS, SELECTION, AND CRITERIA

TABLE R1. Periods of the earlier evaluation (month/day, all dates inclusive)

Period	Training	Calib.	Evaluation segments
Expl. 1 (2013–14)	11/04–12/01	12/02–12/08	12/09–12/22 12/23–01/05 (holidays)
Expl. 2 (2014)	02/03–03/02	03/03–03/09	03/10–03/23 03/24–04/13 04/14–04/27 (Easter)
Held-out (2014)	05/05–06/01	06/02–06/08	A: 06/09–06/20 B: 06/21–06/30

We use three disjoint periods (Table R1), each split into four weeks of training, one week of calibration, and two or three evaluation segments. The two exploratory periods were used to develop the methods, to choose the groups, and to compare candidates; the layout attributes branching and isolation were added after analyses of the misses on these periods. The third period was held out.

Method selection and criteria

Among the candidates (MVP on the uniform and the logarithmic scale, `qr_groups`, and `qr_ogd`), we retained those for which, in all five exploratory segments, no group had a 95% Wilson upper confidence bound below 0.94 and whose mean cost was at most 1.20 times that of the hindsight oracle, and chose the cheapest. This selected `mvp_log`, the only candidate meeting both conditions; `qr_ogd` became the secondary method by the tie-break rule among the candidates meeting the cost condition. The same two conditions are the evaluation criteria on the held-out period. The value 0.94 was in the plan from its first draft, initially as a bound on the point estimate of the worst group; its present form, which counts groups significantly below 0.94, and the cost factor 1.20, i.e., at most 20% more search than the oracle, were set after exploratory results with an earlier, smaller collection of groups and before the final groups were evaluated. With the final groups, the exploratory cost of the selected method was 1.19 times that of the oracle. The selection rule and the criteria were recorded in version control before the held-out period was evaluated.

Main results and additional analyses

The held-out period was evaluated once to obtain the main results (Appendix I-C). All further analyses, listed in Section IV of the supplementary material, were performed after the main results were known; they reuse the held-out period and, where stated, the exploratory periods, and were not used to revise the methods or criteria.[1]

Criteria of the earlier evaluation

For each segment and each group with at least 200 hops, we report the recall with its 95% Wilson interval [1]. We call 0.94 the *bar*. The group criterion requires that no group be significantly below the bar in either held-out segment; the cost criterion requires a mean cost of at most 1.20 times that of the oracle. The bar lies one point below the target because the online methods

[1]The evaluation code initially tabulated a slightly different collection of groups (a first-hop indicator added, the group of all hops omitted). Recomputing all statistics on the 25 groups from the stored hop-level results changed no decision reported here.

aim at recall *equal* to the target, so large groups slightly below it would otherwise be flagged; omitting a multiple-testing correction makes the criterion stricter. Because not rejecting is not evidence of adequate recall, we also report the worst group, the count of groups significantly below 0.95, and the number of evaluation seeds for which the criterion holds. Cost is the mean camera-seconds per hop, relative to the oracle of the same predictor and segment.

B. MARGINAL CALIBRATION MISSES HARD GROUPS

On held-out segment B (Fig. R2 of the supplementary material), split conformal calibration nearly meets the target on average (0.944) but leaves 8 groups below the bar, in identifiable situations: the first hop after a lost track is re-acquired (recall 0.870), hops leaving sites with high branching (0.914) or high isolation (0.912), the first hop of a trip (0.924), and slow exits (0.926). Hops leaving sites with low branching or isolation reach about 0.97: the average is met by over-searching easy hops. Segment A shows the same pattern. The failure is not a matter of tuning: the hindsight oracle, whose single threshold meets the target exactly on the evaluation segment, still leaves 6 and 5 groups below the bar (Table S8). Which attributes matter depends on the layout: on a synthetic layout of the same city, exit speed rather than isolation dominated (Table R13), which suggests targeting many attributes at once rather than one known difficulty.

C. MAIN RESULTS ON THE HELD-OUT PERIOD

`mvp_log`, selected as primary method under the earlier plan, meets both criteria (Table S8): no group is below the bar in either segment (worst groups 0.942 and 0.927), and its cost is 1.183 and 1.171 times that of the oracle, both below 1.20. With a bar on the point estimate of the worst group, the form of the first draft of the plan, it would not pass in segment B. `qr_ogd`, selected as secondary method, has a similar cost (1.189) but leaves one group below the bar in segment B. The two are not statistically distinguishable: their worst groups in segment B, re-acquisition hops in both cases, have overlapping intervals, and a McNemar test on the paired outcomes does not reject equality ($p = 0.18$).

No baseline meets both criteria. The marginal rules (`static`, ACI, `mvp_global`) and the oracle leave 4 to 8 groups below the bar, Mondrian 3 and 6, and `qr_groups` three in segment B. MVP on the uniform scale and the CityTrac-style rule meet the group criterion only at higher cost: 1.45 and 1.93 times the oracle for the former, and 2.59 and 2.41 times for the latter (target 0.95); the latter trusts uncalibrated probabilities and reaches a mean recall of 0.97. With the bar at the target itself, every method except the CityTrac-style rule with target 0.99 (at 7.9 and 7.3 times the oracle cost) leaves groups significantly below 0.95. For `mvp_log`, these shortfalls are at most 0.6 points in segment A and, apart from the re-acquisition group, 0.5 points in segment B, as expected for a method that targets recall equal to the target.

The oracle meets the target only on average. Against two references that meet it in every group in hindsight (Table R14; additional analysis), the cost of `mvp_log` equals that of a group quantile regression fitted on the evaluation segment itself (ratios 1.00 and 0.99), while the rule that gives each hop the lowest hindsight threshold of its groups costs 1.44 and 1.45 times the oracle. Online calibration thus reaches the cost of a group-conditional fit in hindsight, but not its worst-group recall (0.950).

A comparison at equal cost is given in Section IV-C of the supplementary material.

D. REANALYSIS IN THE CONFIRMATION FORMAT

After the confirmation results were known, the held-out period was reanalysed with the targets, estimates, comparators, hindsight references, and decision rules of the confirmation experiment, as the locked plan specified in advance. This is an additional analysis of a period whose results were already known. It is reported in Section V of the supplementary material as a check of consistency between periods; the verdicts of the confirmation period stand.

II. CONFIRMATION EXPERIMENT: FURTHER RESULTS

This appendix reports the results of the confirmation experiment that the main text summarizes. All values are from both blocks pooled and averaged over seeds unless stated otherwise; intervals are the conservative 95% intervals of the taxi and day cluster bootstraps.

Other targets

Table S1 extends Table 3 to targets 0.90, 0.93, 0.97, and 0.98.

All pre-specified comparisons

Table S2 classifies every pre-specified comparison of an online method with a comparator for the three estimates (re-acquisition shortfall, isolation Q4 shortfall, and $\bar{D}$) at the five targets, with 99% simultaneous intervals, and gives for each pair the number of pairs of target points at which one method dominated the other: its 95% upper bound of cost was below the other's 95% lower bound, and its 95% lower bound of worst-group recall above the other's 95% upper bound (Sec. 10.5 of the analysis plan in the supplementary material).

TABLE R2. Parameter values of the experiments.

Item	Symbol	Value	Note
Data and preprocessing			
Bounding box (lon, lat)		[-8.7, -8.55] × [41.1, 41.2]	a trip is kept if at least one GPS point lies inside
Minimum points per trip		3	trips with fewer points are dropped
Maximum speed		50 m/s	any step faster: whole trip dropped
Missing-data trips		dropped	trips flagged in the data set as having missing points
GPS interval / interpolation step		15 s / 1 s	linear interpolation, no map matching
Cameras			
Cameras in the file		239	camera layer of the Porto open-data portal (version retrieved in July 2025)
Tunnel cameras		excluded	cameras flagged as inside a tunnel; cameras without the flag are treated as outside
Merging		complete linkage, 50 m	176 sites
Observation radius		50 m	disk around each site; at most 4 sites queried per point
Hops and groups			
Delay bin / horizon / bins	Δ, $T_{\max}$, B	5 s, 600 s, 120	no re-appearance within $T_{\max}$: exit
Time-of-day edges		0, 6, 10, 16, 20, 24 h	local time of the exit
Exit zone		4 sectors	E, N, W, S sector of the last point in the source disk relative to the site, boundaries at $\pm 45°$
Exit speed		last 4 s in the disk	straight-line distance between the last interpolated point in the source site's disk and the point 4 s earlier (or the first point of the sighting), divided by the elapsed time
Branching		entropy	entropy of the source site's next-site distribution over training non-exit hops
Isolation		distance	distance from the source site to the nearest other site
Quartile edges		training period	speed, branching and isolation, over the training hops; set for each period from its own training period (held-out: 05/05–06/01, 2014, dates inclusive)
Source unseen in training			branching: highest quartile; predictor: flat matrix at ϵ_0
Predictor			
Candidate radius	R_{cand}	2500 m	plus every next site observed in training
Shrinkage	λ	20	toward the parent key
Base probability	ϵ_0	0.0001	prior of the coarsest level
Delay smoothing	σ_g	1 bin	Gaussian along the delay axis
Context keys		source site, exit zone, time of day, exit-speed quartile	backed off to the deepest key seen in training
Feedback			
Feedback fraction	ρ	0.3	random over evaluation hops (evaluation seed)
Resolution time		arrival / $t_{\text{exit}} + T_{\max}$	hit resolved at re-appearance, miss at the end of the horizon
Calibration methods			
ACI	γ_A, W	0.0005, 2000	window initialized with the last 2000 calibration scores; α_t clipped to [-0.5, 1]
Mondrian		time of day × first hop of a track × exit-speed quartile	cells with fewer than 200 calibration hops use the global threshold
qr_groups		level α, log score	intercept + reference-coded indicators of the six attributes; subsample of at most 80,000 calibration hops; score floor 1e-08
qr_ogd	γ, clip	0.005, ± 3	offsets start at 0 (threshold of qr_groups), log scale
MVP (mvp_log)	m, r, ϵ	40, 1000, 1	η = 0.0319 (25 groups), 0.1167 (1 group)
	$a_{\min}$	0.002	warm start: all calibration hops in time order with immediate feedback
Evaluation			
Target / group bar	$1-\alpha$	0.95 / 0.94	groups significantly below the bar: 95% Wilson upper bound
Cost criterion		1.20	mean of A and B, cost / hindsight oracle
Minimum group size		200	smaller groups not judged
Evaluation seeds		0 (main); 0, 1, 2, 3, 4	additional seed runs

Blocks

Fig. R5 repeats Figs. 1 and 2 for each block, and Table R3 gives the between-block differences of $\bar{D}$, with 95% intervals, and of the re-acquisition shortfall, as point estimates.

Partitions not used for calibration

Table S3 gives, for the pre-specified alternative partitions, the paired difference between their largest shortfall and that on the 25 groups, and the shortfall on the intersections I1–I6.

TABLE R3. Difference between the two blocks (autumn minus August) in $\bar{D}$ (pp, with 95% intervals) and in the re-acquisition shortfall (pp, point estimates), per method and target.

Method	0.90	0.93	0.95	0.97	0.98
Worst-group shortfall $\bar{D}$, autumn minus August (pp), 95% interval					
`mvp_log`	−0.8 [−2.7, 0.9]	−1.4 [−2.9, 0.4]	−1.0 [−2.1, 0.6]	−0.6 [−1.6, 0.7]	−0.4 [−1.3, 0.6]
`qr_ogd`	0.1 [−1.6, 1.7]	−0.2 [−1.5, 1.5]	−0.4 [−1.3, 1.0]	−0.1 [−0.7, 0.4]	0.2 [−0.3, 0.7]
`static`	−0.6 [−3.1, 1.2]	−0.7 [−3.5, 1.9]	−1.3 [−3.7, 1.3]	−0.6 [−2.7, 1.5]	−1.2 [−2.8, 0.6]
ACI	−0.4 [−2.6, 1.2]	−1.1 [−3.4, 1.2]	−1.3 [−3.5, 1.0]	−1.1 [−2.9, 0.7]	−0.8 [−2.1, 0.6]
Mondrian	−1.4 [−1.8, −0.8]	−1.1 [−2.4, −0.6]	−0.9 [−2.5, 0.6]	−1.0 [−2.3, 0.7]	−0.8 [−1.9, 0.4]
`qr_groups`	−0.2 [−1.5, 1.3]	−0.5 [−1.9, 1.0]	−0.7 [−1.5, 0.5]	−0.2 [−0.7, 0.3]	0.1 [−0.3, 0.6]
`qr_groups_refit`	0.0 [−1.7, 0.6]	−0.3 [−1.9, 0.5]	−0.3 [−1.2, 0.4]	−0.1 [−0.4, 0.4]	0.1 [−0.2, 0.7]
`qr_groups_refit7`	−0.3 [−2.4, 0.7]	−0.6 [−2.0, 0.6]	−0.3 [−1.3, 0.8]	−0.1 [−0.9, 0.7]	0.2 [−0.5, 0.8]
CityTrac-style	1.5 [−0.7, 3.7]	0.7 [−1.1, 2.7]	0.8 [−0.7, 2.6]	1.4 [0.1, 3.0]	1.4 [0.1, 3.0]
Spatula-style	2.3 [−0.4, 5.0]	2.3 [−0.4, 5.0]	2.3 [−0.4, 5.0]	2.3 [−0.4, 5.0]	2.3 [−0.4, 5.0]
Re-acquisition shortfall, autumn minus August (pp), point estimate					
`mvp_log`	−1.0	−1.7	−1.1	−0.6	−0.4
`qr_ogd`	0.3	0.1	−0.0	0.3	0.7
`static`	−0.6	−0.7	−1.3	−0.6	−1.2
ACI	−0.3	−1.1	−1.3	−1.1	−0.8
Mondrian	−1.0	−1.6	−0.9	−1.0	−0.8
`qr_groups`	0.3	−0.2	−0.3	0.2	0.6
`qr_groups_refit`	−0.6	−1.0	−0.6	0.2	0.5
`qr_groups_refit7`	−1.1	−0.9	−0.4	−0.0	0.3
CityTrac-style	1.5	0.7	0.8	1.4	1.4
Spatula-style	2.3	2.3	2.3	2.3	2.3

Diagnostics, tuning history, and trial run

The diagnostics of the confirmation runs (seed ranges, the ACI variants, the share of hops at $a_{\min}$, and reachable upper bounds) and the tuning history of each method with the trial run before the lock are reported in Tables R23 and R24 of the supplementary material.

A. TRACK CUSTODY (ADDITIONAL ANALYSIS)

This analysis was decided after the confirmation results were known and was not part of the locked plan. Table S4 reports the share of chains whose first k hops were all found, $k = 1, \ldots, 5$, with the definition of Table R19 (cumulative minimum of the open-loop hits).

III. IMPLEMENTATION DETAILS AND AVAILABILITY

TABLE R4. Main notation

Symbol	Meaning
$\mathcal{C}, N$	set and number of camera sites
$\mathcal{R}_c$	observation region of site c
t^{exit}, s	end time and source site of an episode
x	exit context observed at t^{exit}
$\Delta, T_{\max}, \mathcal{B}$	bin width, horizon, and bin index set
$\mathcal{V}$	visibility set of a hop
$\mathcal{S}$	search set (camera–time pairs)
$\hat{p}(c, b \mid x)$	estimated visibility probability
τ, σ	threshold and score of a hop
$1 - \alpha$	target recall
$\mathcal{G}$	collection of (overlapping) groups
ρ	fraction of hops that provide feedback
p_{det}	per-visit detection probability
Method (Section IV)	
$\mathcal{G}(x)$	groups that contain context x
$q, \tau(q)$	calibration index and its threshold
$a_{\min}$	smallest calibration miss level (log scale)
m, B_i	number of buckets and i-th bucket
$V_{G,i}, n_{G,i}$	coverage surplus and count, group G, bucket i
η	MVP learning rate
θ_G, γ	online group offset and its step size

Data

The trajectory file contains 1,710,670 trips of 448 distinct taxi identifiers (the dataset description mentions 442). We keep trips with at least one GPS point inside the bounding box and at least three points, and discard trips flagged as having missing data and trips with any step implying a speed above 50 m/s. The trajectories are available from the UCI Machine Learning Repository under the Creative Commons Attribution 4.0 (CC BY 4.0) license [2]. The camera layer is the dataset ''Sistema de Gestão de Tráfego – CCTV'' of the open-data portal of the Municipality of Porto [3] (last updated February 7, 2025); because the portal was not reachable from outside Portugal, we retrieved the file through the Internet Archive (snapshot of July 9, 2025). It lists 239 cameras with a tunnel flag; we exclude the one camera flagged as inside a tunnel, where GPS positions are unreliable, and treat cameras without the flag as outside tunnels. The current version of the layer, released under the Creative Commons CC0 1.0 Universal public-domain dedication, lists 243 cameras; we kept the version used from the start of the study. The code, the configuration files, and the result summaries will be made available in a public repository upon publication.

Visibility predictor

The predictor conditions on the keys $k_1(x) = (s)$, $k_2(x) = (s, z)$, $k_3(x) = (s, z, h)$, and $k_4(x) = (s, z, h, v)$, where z is the exit zone, h the time-of-day period, and v the exit-speed quartile. For level d and key k, let $n_d(k)$ be the number of non-exit training hops with $k_d(x) = k$, and let

$$M_d(k)[c, b] = \sum_{t:\, k_d(x_t)=k} \mathbb{1}\big[(c, b) \in \mathcal{V}_t\big] \tag{R1}$$

for $c \in \mathcal{C}_s$ and $b \in \mathcal{B}$ be the number of those hops in which the object was visible in site c during bin b. To account for small timing errors, each row of $M_d(k)$ is smoothed along the time axis with a Gaussian kernel of width σ_g bins, giving $\tilde{M}_d(k)$. The estimates are then shrunk toward the next coarser level:

$$P_1(k) = \frac{\tilde{M}_1(k) + \lambda\, \epsilon_0}{n_1(k) + \lambda}, \tag{R2}$$

$$P_d(k) = \frac{\tilde{M}_d(k) + \lambda\, P_{d-1}(k')}{n_d(k) + \lambda}, \qquad d = 2, 3, 4, \tag{R3}$$

where k' is the parent key of k, $\lambda > 0$ controls the strength of shrinkage, ϵ_0 is a small floor, and all entries are clipped to $[0, 1]$. For a hop with context x, the predictor uses the finest level whose key was observed in training, $\hat{p}(c, b \mid x) = P_{d^*}(k_{d^*}(x))[c, b]$; a source site never observed in training receives the flat value ϵ_0. Entries estimate the probability that a pair belongs to the visibility set, not that the object is first seen there, and need not sum to one. For each key, the entries are stored in sorted order, so the cost of any threshold follows from a binary search. The calibration layer adds $O(|\mathcal{G}(x)|\, m)$ operations per hop.

Uniform scale (Section IV of version 7c55fbc)

The original *uniform* scale corresponds to

$$\tau_{\mathrm{uni}}(q) = \hat{F}^{-1}(1 - q), \tag{R4}$$

so that q is a coverage level on the calibration data.

Why a logarithmic scale

On the uniform scale, a bucket spans $1/m = 0.025$ in coverage level, so near the target one bucket changes the miss level by up to a factor of two, and $q = 1$ searches all candidates. Since cost grows faster than linearly as the miss level falls, MVP's randomization over this coarse grid raises the average cost. On the logarithmic scale, one bucket changes the miss level by the constant factor $a_{\min}^{1/m} \approx 0.856$, which refines the grid where the target lies, and the most permissive threshold is bounded at $\tau_{\min} = \hat{F}^{-1}(a_{\min})$. the ablation in Section IV of the supplementary material separates the two effects. The bound has a cost: hops with $\sigma < \tau_{\min}$ are missed whatever MVP does, so a group can reach its target only if

$$\Pr\big[\sigma < \tau_{\min} \,\big|\, x \in G\big] < \alpha. \tag{R5}$$

This holds on the calibration data, where the left-hand side is about $a_{\min}$, but can fail when the predictor is misspecified (Section IV of the supplementary material).

MVP procedure

Algorithm R1 states the procedure of Section IV-D1, including the warm start and the handling of delayed, partial feedback. The learning rate is $\eta = \sqrt{\ln(|\mathcal{G}|m)/(2K_\epsilon|\mathcal{G}|m)}$ with $K_\epsilon = \sum_{n\geq 0} f(n)^{-2}$.

Table R2 lists all parameter values.

Algorithm R1 MVP threshold calibration with delayed, partial feedback

Require: calibration scores and contexts in time order; parameterization $\tau(\cdot)$ from (R4) or (6); m, r, η

$V_{G,i} \leftarrow 0$, $n_{G,i} \leftarrow 0$ for all $G \in \mathcal{G}$, $i \in [m]$
for each calibration hop in time order **do** ▷ warm start
    choose q as in lines 9–14; update (V, n) immediately with its outcome
end for
for each test hop t in order of t_t^{exit} **do**
    for each feedback hop j resolved by t_t^{exit} **do**
        $i \leftarrow$ bucket of q_j; for all $G \ni x_j$: $n_{G,i}$ += 1, $V_{G,i}$ += $\mathbb{1}[\sigma_j \geq \tau(q_j)] - (1-\alpha)$
    end for
    compute $C_i(x_t)$ for $i \in [m]$ by (8)
    if $C_i > 0$ for all i **then** $q_t \leftarrow 0$
    else if $C_i < 0$ for all i **then** $q_t \leftarrow 1$
    else
        $i^* \leftarrow$ first index with $C_{i^*} C_{i^*+1} \leq 0$; $p \leftarrow |C_{i^*+1}|/(|C_{i^*}| + |C_{i^*+1}|)$
        $q_t \leftarrow i^*/m - 1/(rm)$ w.p. p, else $q_t \leftarrow i^*/m$
    end if
    search $\mathcal{S}_{\tau(q_t)}(x_t)$
    if hop t provides feedback **then**
        schedule its outcome at its arrival time if hit, else at $t_t^{\text{exit}} + T_{\max}$
    end if
end for

TABLE R5. Worst group of each method on the held-out period (evaluation seed 0): group, recall, number of hops *n*, and 95% Wilson interval. †Added after the main results were known.

	Segment A				Segment B			
Method	Group	Recall	*n*	95% CI	Group	Recall	*n*	95% CI
static	Re-acquired	0.907	1,054	[0.888, 0.923]	Re-acquired	0.870	846	[0.846, 0.891]
ACI	Re-acquired	0.904	1,054	[0.885, 0.921]	Re-acquired	0.881	846	[0.857, 0.901]
Mondrian	Branching Q4 (high)	0.919	66,829	[0.917, 0.921]	Re-acquired	0.885	846	[0.862, 0.905]
Spatula-style	Re-acquired	0.825	1,054	[0.801, 0.847]	Re-acquired	0.824	846	[0.797, 0.848]
CityTrac-style (0.95)†	Speed Q1 (slow)	0.945	63,228	[0.943, 0.947]	Re-acquired	0.933	846	[0.914, 0.948]
CityTrac-style (0.99)†	Speed Q1 (slow)	0.963	63,228	[0.961, 0.964]	Re-acquired	0.956	846	[0.940, 0.968]
qr_groups	Re-acquired	0.942	1,054	[0.926, 0.955]	Re-acquired	0.915	846	[0.894, 0.932]
mvp_groups	Re-acquired	0.942	1,054	[0.926, 0.955]	Re-acquired	0.931	846	[0.912, 0.947]
mvp_global	Re-acquired	0.906	1,054	[0.887, 0.922]	Re-acquired	0.877	846	[0.853, 0.898]
qr_ogd	Re-acquired	0.943	1,054	[0.927, 0.956]	Re-acquired	0.921	846	[0.901, 0.937]
mvp_log	Re-acquired	0.942	1,054	[0.926, 0.955]	Re-acquired	0.927	846	[0.907, 0.942]
hindsight oracle	Re-acquired	0.909	1,054	[0.890, 0.925]	Re-acquired	0.876	846	[0.852, 0.896]

FROM THE SUPPLEMENTARY MATERIAL OF VERSION 7C55FBC

This document collects results that support statements of the paper but are not needed to follow its argument. Section IV reports additional analyses of the held-out period of the earlier evaluation, performed after its main results were known; Section V reanalyses that period in the format of the confirmation experiment; and Section VI gives per-block curves, diagnostics, and the tuning history and trial run of the confirmation experiment. Fig. R1 and Table R5 complement the main results on the held-out period. Tables R6 and R7 give the full outage results on the exploratory periods, Table R8 covers the first 48 h after the outage start on the held-out period, Table R9 decomposes the hops affected by the outage, Table R10 reports the share of hops below the most permissive threshold of `mvp_log`, and Tables R11 and R12 compare `mvp_log` with the hindsight oracle of the same outage scenario, by group and by hop type.

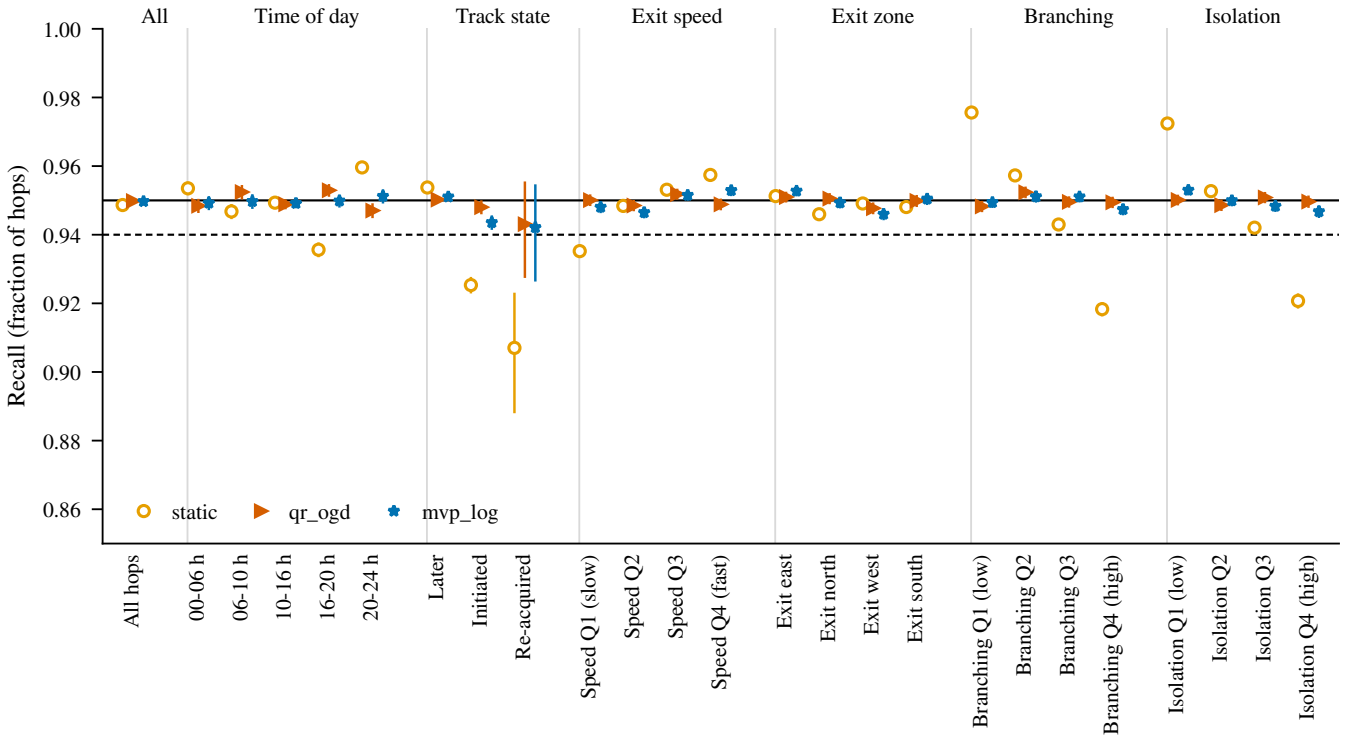


FIGURE R1. Recall per group on held-out segment A, in the format of Fig. R2.

IV. ADDITIONAL ANALYSES ON THE HELD-OUT PERIOD

All analyses in this section were performed on the held-out period of Appendix I after its results were known. None of them includes the refitted comparators of the confirmation experiment; in particular, the outage analyses do not compare online

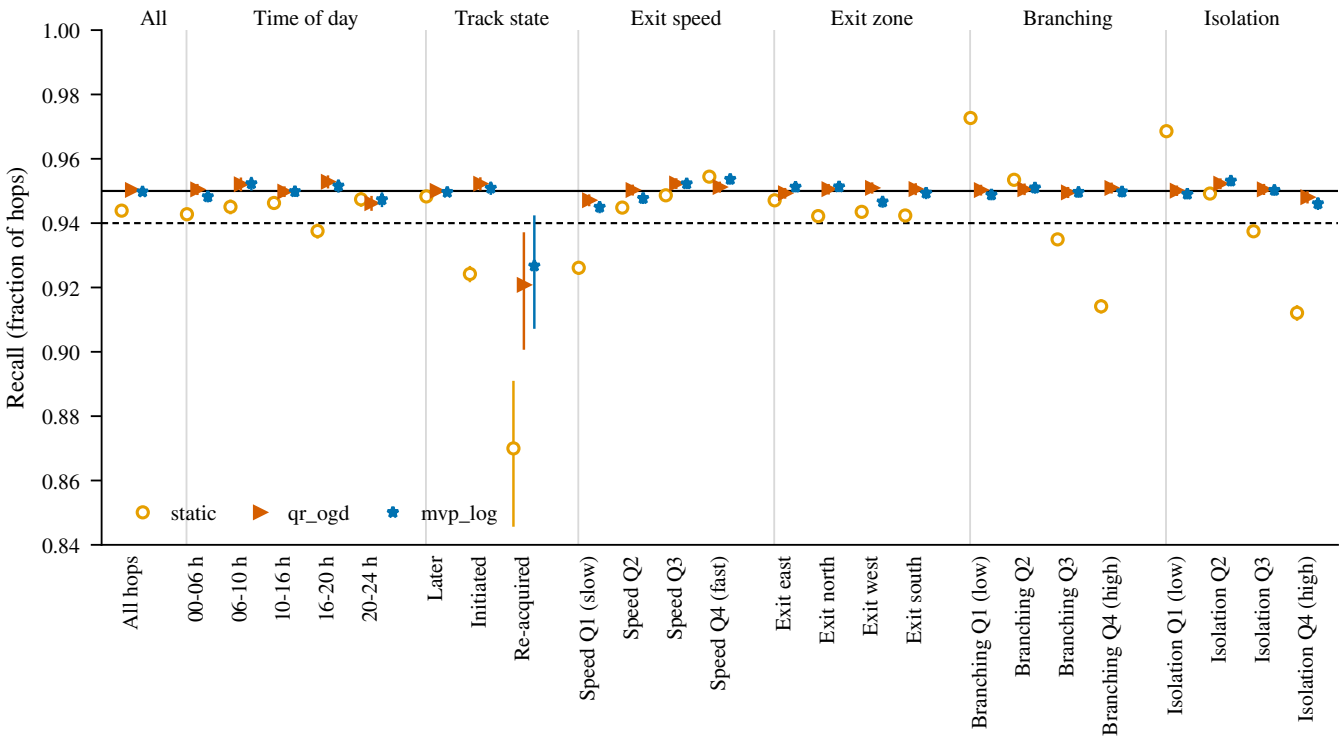


FIGURE R2. Recall per group on held-out segment B for split conformal calibration (`static`), `qr_ogd`, and `mvp_log`, with 95% Wilson intervals. Groups are ordered by attribute; isolation is the distance from the source site to the nearest other site. Solid line: target 0.95; dashed line: bar 0.94. Segment A is shown in Fig. R1.

TABLE R6. Camera outages on exploratory period 1, in the format of Table R20, from 48 h after the outage start, which is the start of the period's second evaluation segment, to the end of its last evaluation segment. In this period, the second segment is the Christmas and New Year holiday period, so the outage starts together with the holidays.

		static			ACI			qr_ogd			mvp_log		
Outage	Pred.	Recall	<0.94	Cost	Recall	<0.94	Cost	Recall	<0.94	Cost	Recall	<0.94	Cost
Random 10%	base	0.871	24.8	0.89	0.950	5.0	28.13	0.938	11.4	10.88	0.936	12.8	16.27
	aware	0.928	19.4	1.01	0.950	4.8	1.71	0.950	1.2	1.87	0.951	0.6	2.07
Random 20%	base	0.823	25.0	0.81	0.950	4.0	52.08	0.924	20.4	21.70	0.926	18.0	24.44
	aware	0.904	24.0	0.99	0.950	5.0	2.92	0.950	1.6	3.17	0.948	1.4	3.52
Random 30%	base	0.763	25.0	0.75	0.950	5.2	65.48	0.912	23.4	32.94	0.902	20.4	29.35
	aware	0.880	24.8	1.00	0.950	3.6	4.09	0.950	2.2	4.10	0.948	0.8	4.54
Region (20 sites)	base	0.937	11	0.95	0.950	6	1.49	0.949	3	1.57	0.947	4	3.52
	aware	0.938	10	0.96	0.950	6	1.46	0.950	3	1.60	0.951	1	2.11

updating with refitting, and we do not use them as evidence for online updating.

A. ROBUSTNESS

Evaluation seeds

Over 5 evaluation seeds, including that of the main results, the group criterion holds for `mvp_log` in 3 and for `qr_ogd` in 2, and the cost criterion in all (Table R15). Every failure is a single group, the re-acquisition hops in segment B (recall 0.909 to 0.928 for `mvp_log`). This group contains about 0.4 % of the hops, so few feedback hops fall into it; it is the persistent weak point of both methods.

Observation radius and horizon

Repeating the evaluation with observation radii of 30 and 75 m and horizons of 300 and 900 s, with hops, groups, and models rebuilt (Table R16), marginal calibration fails in every setting (4 to 8 groups below the bar). The multigroup methods meet the group criterion in most seeds of most settings, at mean costs of 1.06 to 1.23 times the oracle, but not uniformly: with the 900-s horizon, `mvp_log` fails in all five seeds, on the re-acquisition group, whereas `qr_ogd` meets the group criterion in four at a

TABLE R7. Camera outages on exploratory period 2, in the format of Table R20, from 48 h after the outage start to the end of the period's last evaluation segment.

		static			ACI			qr_ogd			mvp_log		
Outage	Pred.	Recall	<0.94	Cost	Recall	<0.94	Cost	Recall	<0.94	Cost	Recall	<0.94	Cost
Random 10%	base	0.868	25.0	0.86	0.950	5.2	33.84	0.941	8.6	16.40	0.933	14.8	15.44
	aware	0.923	22.2	0.98	0.950	5.2	1.92	0.950	0.6	2.44	0.950	0.2	2.48
Random 20%	base	0.821	25.0	0.79	0.950	5.4	54.59	0.931	15.0	30.35	0.921	20.4	22.04
	aware	0.903	24.6	0.96	0.950	4.2	2.92	0.950	0.8	3.68	0.949	0.2	3.81
Random 30%	base	0.759	25.0	0.72	0.950	5.2	72.29	0.920	18.0	44.23	0.890	22.8	25.11
	aware	0.876	25.0	0.96	0.950	4.0	4.82	0.949	0.4	5.07	0.949	0.0	5.26
Region (20 sites)	base	0.933	17	0.92	0.950	6	1.61	0.949	3	1.97	0.944	5	3.04
	aware	0.934	17	0.93	0.950	6	1.56	0.949	2	2.03	0.949	0	2.82

TABLE R8. Camera outages on the held-out period during the first 48 h after the outage start, in the format of Table R20.

		static			ACI			qr_ogd			mvp_log		
Outage	Pred.	Recall	<0.94	Cost	Recall	<0.94	Cost	Recall	<0.94	Cost	Recall	<0.94	Cost
Random 10%	base	0.869	23.8	0.94	0.950	4.2	42.89	0.912	20.4	3.65	0.896	23.4	5.63
	aware	0.928	17.8	1.07	0.949	3.0	1.83	0.944	3.6	1.77	0.929	16.0	1.48
Random 20%	base	0.828	24.0	0.85	0.951	3.8	55.89	0.887	23.8	5.15	0.874	24.0	8.99
	aware	0.909	22.0	1.04	0.951	1.4	2.81	0.940	5.2	2.34	0.913	21.2	1.63
Random 30%	base	0.767	24.0	0.79	0.950	3.8	84.06	0.845	24.0	6.43	0.835	24.0	12.37
	aware	0.884	23.6	1.05	0.950	2.8	4.75	0.933	12.0	2.87	0.897	23.4	1.92
Region (20 sites)	base	0.938	8	1.01	0.948	5	1.44	0.944	4	1.38	0.937	8	1.23
	aware	0.939	8	1.02	0.947	5	1.40	0.944	4	1.42	0.938	5	1.26

cost of 1.19 to 1.23 times the oracle, and both criteria in only one. We did not investigate why. Neither instance dominates the other across settings.

Threshold scale

On the exploratory periods (Table R17), the logarithmic scale reduces the cost from 1.92 to 1.18 times the oracle; the finer grid accounts for 49.9 % and 74.2 % of the reduction in the two periods and the bound $a_{\min}$ for the rest. With the logarithmic grid, removing the bound changes the cost only to 1.19 but, like $a_{\min} = 0.005$, leads to one exploratory segment with a group below the bar.

Detection target (Section IV of version 7c55fbc)

To target a true recall of $1 - \alpha$, the observed target is set to

$$1 - \alpha_{\text{obs}} = (1 - \alpha)\, p_{\text{det}}, \tag{R6}$$

which requires an estimate of p_{det} for the deployed detector.

Feedback and detection

The results hardly depend on the feedback model (Table R18): with 10 % to 100 % delayed feedback, at most one group (re-acquisition) falls below the bar, at 1.14 to 1.18 times the oracle cost. Imperfect detection matters: with detection probabilities of 0.95 and 0.90 and the target unchanged, the cost of `mvp_log` rises to 86.22 and 136.98 times the error-free oracle, whereas the adjusted target (R6) restores a true recall of about 0.948 at 1.17 times for $p_{\text{det}} = 0.95$, and of 0.944 for $p_{\text{det}} = 0.90$, with two groups below the bar in one segment. Detection failures are assumed independent.

Track custody

A track survives only if consecutive hops are all hit (Table R19). After five hops, 80 to 82 % of the chains survive under every method, slightly more than independence would give, because consecutive hits are positively correlated. Group-conditional calibration does not raise custody in general: over all chains, the multigroup methods keep slightly fewer chains than the oracle and ACI, because on later hops, which make up most of a chain, their recall is 0.4 to 0.5 points lower (Table R14, lower part). For chains that start with a re-acquisition, `mvp_log` raises the first-hop recall (0.927 against 0.876 in segment B), but over five hops the effect is mixed. A custody target thus requires a higher per-hop target, which we did not evaluate.

TABLE R9. Hops affected by camera outages on held-out segment B (original predictor's hop set without outage as reference). Removed: hops whose source site failed, and hops that became exits. Rerouted: hops whose object re-appears at a later site after skipping one or more failed sites. Recall of rerouted hops for `mvp_log` with the original and the outage-aware predictor. Random outages: one of the five draws of failed sites, the one used in the first outage experiments; Region: an earlier local outage of six sites.

		Removed (%)		Rerouted (%)		Recall, rerouted	
Outage	Sites off	Source off	To exit	Skip 1	Skip ≥2	Base	Outage-aware
Random 10%	18	11.7	1.5	6.3	1.8	0.275	0.793
Random 20%	35	20.3	4.1	10.6	1.4	0.580	0.811
Random 30%	53	26.4	5.0	7.2	3.5	0.485	0.738
Region (6 sites)	6	4.6	0.4	0.6	0.7	0.086	0.506

TABLE R10. Share of hops whose score lies below the most permissive threshold $\tau_{\min}$ of `mvp_log`, by group, from 48 h after the outage start (held-out period, mean over five draws of failed sites). †Above $\alpha = 0.05$, i.e., the target is infeasible for this group (Section IV-D1).

	Random 10%		Random 30%	
Group	Base	Outage-aware	Base	Outage-aware
All hops	0.050	0.001	0.109†	0.001
00-06 h	0.056†	0.001	0.119†	0.001
06-10 h	0.052†	0.001	0.112†	0.001
10-16 h	0.050	0.001	0.111†	0.001
16-20 h	0.047	0.001	0.102†	0.001
20-24 h	0.043	0.001	0.099†	0.001
Later	0.051†	0.001	0.105†	0.001
Initiated	0.046	0.001	0.123†	0.002
Re-acquired	0.062†	0.001	0.123†	0.004
Speed Q1 (slow)	0.033	0.001	0.096†	0.001
Speed Q2	0.038	0.001	0.098†	0.001
Speed Q3	0.049	0.001	0.115†	0.001
Speed Q4 (fast)	0.084†	0.001	0.131†	0.001
Exit east	0.037	0.001	0.098†	0.001
Exit north	0.059†	0.001	0.108†	0.001
Exit west	0.064†	0.001	0.136†	0.002
Exit south	0.040	0.001	0.098†	0.001
Branching Q1 (low)	0.054†	0.001	0.116†	0.001
Branching Q2	0.040	0.001	0.101†	0.001
Branching Q3	0.052†	0.001	0.112†	0.003
Branching Q4 (high)	0.052†	0.002	0.106†	0.001
Isolation Q1 (low)	0.050†	0.001	0.091†	0.001
Isolation Q2	0.032	0.001	0.088†	0.001
Isolation Q3	0.062†	0.002	0.153†	0.001
Isolation Q4 (high)	0.061†	0.002	0.123†	0.002

B. CAMERA OUTAGES

From the start of segment B, a random 10%, 20%, or 30% of the sites, or the 20 sites around the busiest site, stop operating (Fig. R3, Table R20; additional analysis).

Original predictor

No calibration rule maintains group recall at a reasonable cost. Under the 10% outage, `mvp_log` reaches 0.932 with 14.4 groups below the bar at 14.87 times the cost of the oracle without outage; ACI keeps the mean at 0.949 only at 34.19 times that cost; `static` falls to 0.862. The cause is the predictor: objects re-appear later and farther away, at pairs never observed in training, and `mvp_log` recalls these rerouted hops at only 0.275 (Table R9). The bound $a_{\min}$ turns this into a ceiling: the share of hops below $\tau_{\min}$ rises to 0.050 and 0.109 under the 10% and 30% outages and exceeds α in several groups, violating (R5) (Table R10); meanwhile, the share of hops assigned to the most permissive threshold rises from zero to about one half. Lowering $a_{\min}$ helps only at 21 to 38 times the cost of the oracle without outage. In hindsight, the bound offers little in stationary conditions and limits recovery under misspecification.

TABLE R11. Recall per group of `mvp_log` and of the hindsight oracle of the same scenario, and the cost of `mvp_log` relative to that oracle, under a random outage of 10% of the sites with the outage-aware predictor (held-out period, from 48 h after the outage start; one of the five draws of failed sites, the one used in the first outage experiments).

		Recall		
Group	n	mvp_log	Oracle	Cost ratio
All hops	167,979	0.947	0.950	1.54
00-06 h	37,293	0.943	0.949	1.82
06-10 h	26,930	0.948	0.945	1.79
10-16 h	48,306	0.950	0.951	1.48
16-20 h	30,140	0.949	0.949	1.30
20-24 h	25,310	0.944	0.956	1.41
Later	134,749	0.947	0.952	1.42
Initiated	32,517	0.951	0.945	1.89
Re-acquired	713	0.913	0.853	5.50
Speed Q1 (slow)	41,581	0.948	0.955	1.08
Speed Q2	45,841	0.951	0.966	0.92
Speed Q3	43,534	0.945	0.962	1.13
Speed Q4 (fast)	37,023	0.945	0.910	3.99
Exit east	46,714	0.952	0.965	0.98
Exit north	44,111	0.941	0.942	1.60
Exit west	39,162	0.948	0.925	2.83
Exit south	37,992	0.948	0.967	0.82
Branching Q1 (low)	43,353	0.940	0.983	0.28
Branching Q2	40,090	0.949	0.934	2.19
Branching Q3	44,233	0.950	0.937	2.16
Branching Q4 (high)	40,303	0.950	0.945	1.44
Isolation Q1 (low)	49,550	0.942	0.982	0.34
Isolation Q2	46,136	0.955	0.973	0.58
Isolation Q3	38,376	0.946	0.944	1.58
Isolation Q4 (high)	33,917	0.945	0.879	4.23

TABLE R12. Extra cost of `mvp_log` relative to the hindsight oracle of the same scenario, by hop type, under a random outage of 10% of the sites with the outage-aware predictor (held-out period, from 48 h after the outage start; one of the five draws of failed sites, the one used in the first outage experiments). Share of hops, recall of both rules, and share of the extra cost.

Hop type	Hops	Recall		Extra
	(%)	mvp_log	Oracle	cost (%)
Not rerouted	90.5	0.962	0.972	15.1
Rerouted, 1 failed site skipped	7.2	0.882	0.927	16.5
Rerouted, $\geq$2 failed sites skipped	2.1	0.574	0.083	68.2
New (source episode changed)	0.1	0.293	0.992	-0.5
Re-acquisition caused by the outage	0.1	0.919	0.892	0.7
All	100.0	0.947	0.950	100.0

Outage-aware predictor

Replacing the predictor by one that composes transitions through failed sites, developed after these results were known and replicated on the exploratory periods (Table R21), restores most of the recall with every calibration layer unchanged; the share of hops below $\tau_{\min}$ falls to 0.001 under the 10% outage. After the first 48 h, `mvp_log` reaches 0.944 to 0.945 with 2.4 to 4.6 groups below the bar, and `qr_ogd` 0.948 to 0.949 with 1.8 to 2.4, at 2.30 to 4.22 times the cost of the oracle without outage; the extra cost of `mvp_log` is concentrated on hops that skipped two or more failed sites (Tables R11 and R12). The update does not settle every case: when adjacent sites fail together, many rerouted hops skip several of them, and `mvp_log` still leaves 2 groups below the bar under the local outage. Once the predictor is updated, marginal ACI leaves only 2.8 to 4.0 groups below the bar, so the advantage of multigroup calibration is smaller than in stationary conditions.

The first 48 hours

All methods fall short right after the outage. With the outage-aware predictor, ACI and `qr_ogd`, whose states move by constant steps, stay at or above a mean recall of 0.94 for 24 h from 6 h after the start, `mvp_log` only from 78 h (Table R8). Diagnostic

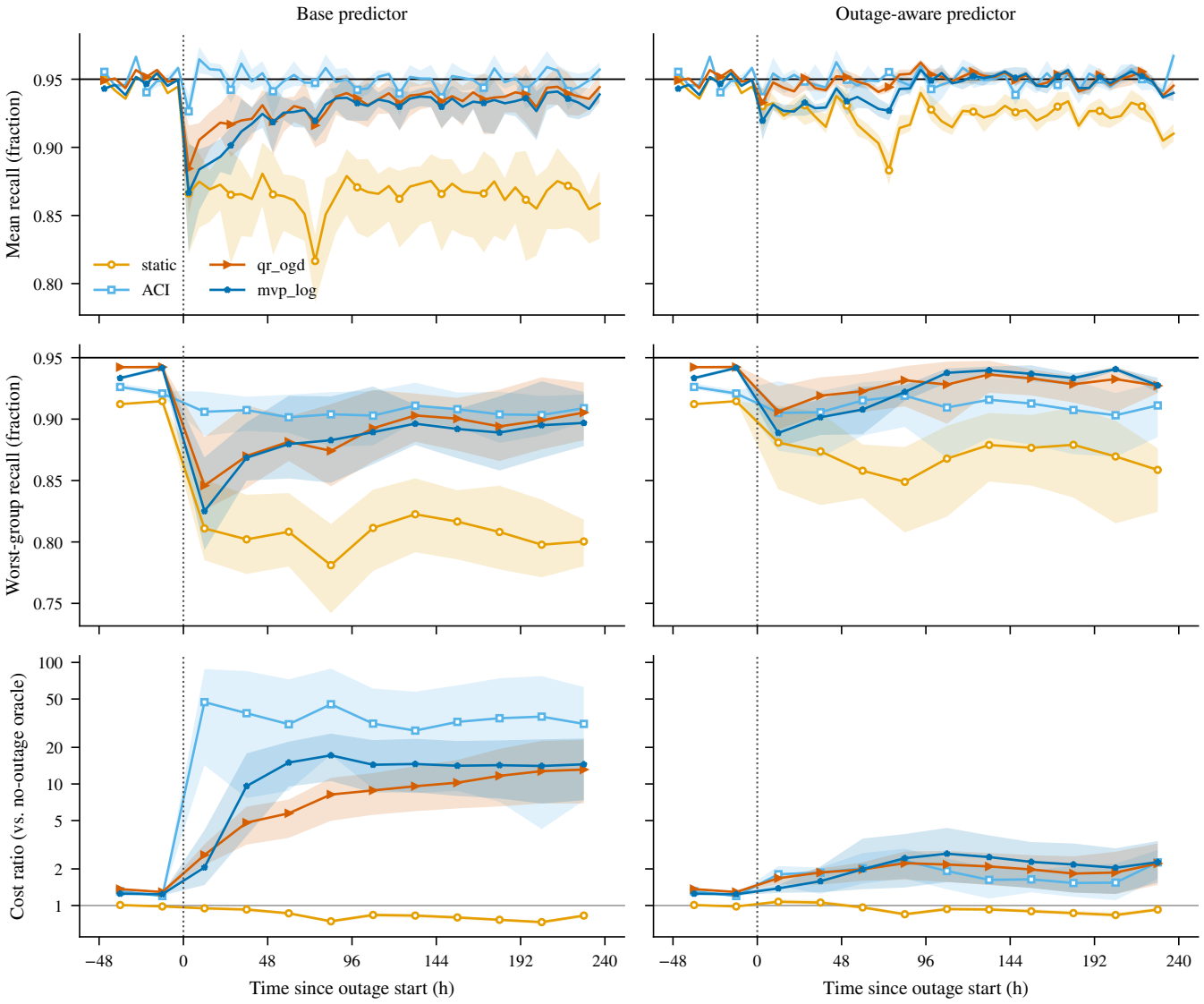


FIGURE R3. **Recall and cost over time under a random outage of 10% of the sites on the held-out period (additional analysis). Lines: mean over five draws of failed sites; bands: minimum to maximum. Top: mean recall (6-h bins). Middle: worst-group recall (24-h bins, groups with at least 200 hops). Bottom: cost relative to the mean cost of the hindsight oracle without outage in the same segment (24-h bins, logarithmic axis). Left: original predictor; right: outage-aware predictor.**

runs support the mechanism of Section IV-E: removing the warm start shortened the recovery of MVP with a single group from 30 h to 12 h. The slow transient is a limitation of `mvp_log`. Under structural change, the predictor must thus be updated; recalibration alone cannot compensate for a predictor that does not know where objects re-appear.

All results in this section are additional analyses, performed after the main results were known.

List of additional analyses

(i) *CityTrac-style baseline*: added to the compared methods and evaluated on the held-out period. (ii) *Evaluation seeds*: the held-out evaluation run with five seeds, including that of the main results, which change the feedback sample, the randomization of MVP, and the calibration subsample of `qr_groups`. (iii) *Cost-matched comparison* (Section IV-C). (iv) *Sensitivity* to the feedback fraction (10 % to 100 %), to feedback delay, and to imperfect detection, with detection probabilities between 0.90 and 0.95. (v) *Scale ablation* on the exploratory periods, separating the effect of the logarithmic grid from that of the bound $a_{\min}$. (vi) *Camera outages*: from the start of the second evaluation segment, some sites stop operating; sighting data of failed sites are removed and hops are rebuilt, so that objects re-appear at the next operating site. We first ran outages on the held-out period only (a random 10 %, 20 %, or 30 % of the sites drawn once, and a local outage of six sites), followed by diagnostics of the adaptation delay, including an ablation of the warm start, and a decomposition of the hops lost or rerouted by the outage. We

then ran random outages drawn with five seeds and a local outage of the 20 sites within a fixed radius of the busiest site on all three periods. Rebuilding the hops changes the order in which hops with the same exit time are processed, and hence the feedback sample, slightly; before the outage, this changes the cost of `mvp_log` by 0.01 % to 2.8 % across the outage runs. (vii) *Outage-aware predictor*: a variant of the predictor that composes transitions through failed sites. We developed it after seeing the outage results on the held-out period; it is therefore an extension informed by those results, and we replicate its evaluation on the two exploratory periods. (viii) *Diagnostics of the bound* $a_{\min}$: the recall ceiling it imposes under outages, the nature of the pairs below $\tau_{\min}$, and a per-group comparison of cost with the hindsight oracle. (ix) *Group-conditional hindsight references*: two rules fitted in hindsight on each held-out segment to meet the target in every group, used as cost references (Section I-C). (x) *Track custody*: the share of chains whose first k hops are all hit. (xi) *Geometry and horizon*: the held-out evaluation repeated, with five evaluation seeds, for observation radii of 30 and 75 m and horizons $T_{\max}$ of 300 and 900 s, with all hops, groups, and models rebuilt for each setting.

Tables

Table R15 lists the results of the two online multigroup methods for each evaluation seed (Section IV-A). Table R14 compares them with two references that meet the target in every group in hindsight (Section I-C), and Table R19 reports track custody over several hops (Section IV-A). Table R16 repeats the held-out evaluation for other observation radii and horizons (Section IV-A). Table R13 compares the attributes that matter on a synthetic and on the real camera layout (Section I-B), and Table R21 replicates the outage experiments with the outage-aware predictor on the exploratory periods (Section IV-B).

TABLE R13. Spread of recall across the levels of each attribute under split conformal calibration (maximum minus minimum over levels, percentage points), on the two exploratory periods, for a synthetic layout of 130 sites placed by traffic density and for the real CCTV layout (additional analysis). The two layouts produce different hops.

	Synthetic layout		Real CCTV layout	
Attribute	Period 1	Period 2	Period 1	Period 2
Time of day	1.1	1.7	1.4	1.6
Track state	6.1	6.5	6.5	5.5
Exit speed	8.2	8.0	3.2	2.6
Exit zone	2.4	2.2	0.5	1.0
Branching	6.4	7.1	5.2	5.3
Isolation	1.9	2.0	5.4	4.5

C. COST-MATCHED COMPARISON

At the cost of `mvp_log` (Fig. R4; additional analysis, one evaluation seed), the worst group of `static` reaches 0.917 and 0.884, and that of the CityTrac-style rule 0.888 and 0.874, against 0.942 and 0.927. To leave no group below the bar, `static` needs 1.69 and 2.29 times the oracle cost, and the CityTrac-style rule 2.59 and 2.41. The price is in the mean: at equal cost, `static` achieves a mean recall higher by 0.7 and 0.6 points, because group-conditional calibration moves search effort from easy to hard hops.

V. REANALYSIS OF THE HELD-OUT PERIOD IN THE CONFIRMATION FORMAT

After the confirmation results were known, the held-out period of the earlier evaluation (Appendix I) was reanalysed with the targets, estimates, comparators, hindsight references, and decision rules of the confirmation experiment, as the locked plan specified in advance (Sec. 12 of the analysis plan). The earlier per-hop results were reproduced exactly. The reanalysis is an additional analysis of a period whose results were already known: it checks consistency between periods and changes no verdict of the confirmation period.

Table R22 lists the verdicts on both periods. The verdicts on group against marginal calibration (S1) and on online updating against daily refitting (S2) were the same on both periods. Two verdicts differed. Against `qr_groups` (S1 addition), `mvp_log` was not distinguished at 3 of five targets, whereas it was worse at 3 on the confirmation period, and `qr_ogd` improved on `qr_groups` at 5 of five targets, whereas it was not distinguished at 5 on the confirmation period. In the rarest group (S4), the 95% lower bound of the re-acquisition shortfall was above zero at a majority of targets on the held-out period for both online methods, whereas on the confirmation period this held for `mvp_log` only.

The ordering among the group methods therefore changed between the two periods, and so did the verdict on the rarest group, while their difference from marginal calibration did not. A change of ordering is what differences within the variability between periods would produce; it does not by itself separate sampling variability from a genuine dependence on the period, and neither period supports a general ranking of the group methods. Following the plan, the main text reports the verdicts of the confirmation period.

TABLE R14. Group-conditional references on the held-out period (evaluation seed 0). (a) Group-minimum rule: each hop uses the lowest of the hindsight thresholds of the groups that contain it. (b) Hindsight group QR: the quantile regression of `qr_groups` fitted on the evaluation segment itself. Cost in camera-seconds per hop, and relative to the hindsight oracle and to rule (a). The lower part reports the later hops only, with costs relative to the hindsight oracle on the same hops.

Seg.	Reference / method	Recall	Worst	Worst group	<0.94	<0.95	Cost	×oracle	×(a)
A	Hindsight oracle (global)	0.950	0.909	Re-acquired	6	9	279	1.00	0.70
	Group-minimum rule (a)	0.964	0.950	Isolation Q4 (high)	0	0	400	1.44	1.00
	Hindsight group QR (b)	0.950	0.950	Exit south	0	0	330	1.18	0.82
	qr_ogd	0.950	0.943	Re-acquired	0	3	334	1.20	0.83
	mvp_log	0.950	0.942	Re-acquired	0	7	330	1.18	0.82
B	Hindsight oracle (global)	0.950	0.876	Re-acquired	5	8	301	1.00	0.69
	Group-minimum rule (a)	0.964	0.950	Re-acquired	0	0	436	1.45	1.00
	Hindsight group QR (b)	0.950	0.950	00-06 h	0	0	357	1.18	0.82
	qr_ogd	0.950	0.921	Re-acquired	1	4	356	1.18	0.82
	mvp_log	0.950	0.927	Re-acquired	0	7	353	1.17	0.81
Later hops only									
A	Hindsight oracle (global)	0.955	–		–	–	270	1.00	–
	ACI	0.955	–		–	–	290	1.07	–
	qr_ogd	0.950	–		–	–	296	1.09	–
	mvp_log	0.951	–		–	–	297	1.10	–
B	Hindsight oracle (global)	0.954	–		–	–	292	1.00	–
	ACI	0.955	–		–	–	325	1.11	–
	qr_ogd	0.950	–		–	–	318	1.09	–
	mvp_log	0.950	–		–	–	315	1.08	–

TABLE R15. Results of `mvp_log` and `qr_ogd` on the held-out period for each evaluation seed (seed 0: main results; seeds 1–4: additional analysis). Group: worst group; Worst: its recall; <0.94: groups below the bar; Cost: mean cost relative to the hindsight oracle over both segments.

	mvp_log							qr_ogd						
	Segment A			Segment B				Segment A			Segment B			
Seed	Group	Worst	<0.94	Group	Worst	<0.94	Cost	Group	Worst	<0.94	Group	Worst	<0.94	Cost
0	Re-acquired	0.942	0	Re-acquired	0.927	0	1.18	Re-acquired	0.943	0	Re-acquired	0.921	1	1.19
1	Initiated	0.944	0	Re-acquired	0.928	0	1.15	Initiated	0.947	0	Re-acquired	0.942	0	1.18
2	Re-acquired	0.931	0	Re-acquired	0.915	1	1.15	Re-acquired	0.939	0	Re-acquired	0.916	1	1.19
3	Re-acquired	0.935	0	Re-acquired	0.928	0	1.15	00-06 h	0.947	0	Re-acquired	0.943	0	1.20
4	Re-acquired	0.927	0	Re-acquired	0.909	1	1.14	Re-acquired	0.940	0	Re-acquired	0.917	1	1.17

VI. CONFIRMATION EXPERIMENT: DIAGNOSTICS AND TUNING HISTORY

Diagnostics

Table R23 reports, for the confirmation period (Appendix II), the range of $\bar{D}$ and of the re-acquisition shortfall over the 5 seeds, three variants of ACI, the share of hops at which `mvp_log` searched at the floor $a_{\min}$ at target 0.98, and the reachable upper bound of recall in the two groups fixed in advance. Of the ACI variants, two differ only in the step size; the third is a group-wise ACI over time of day and first hop (10 cells), which was not a pre-specified comparator. ACI with separate states per Mondrian cell was not run; a group-wise ACI over the 25 groups is among the additional baselines of Section VI-C. Results of `qr_ogd` depend on the seed: at target 0.90, its range over seeds was 0.79 of the width of its interval.

Trial run and tuning history

Before the lock, the full pipeline was run once on exploratory period 1 to check it (Sec. 13.8 of the analysis plan). The check of detection times failed in that run, and the recording code was corrected (commit `b5e10e6`); no method, comparator, or decision rule was changed after the trial run, and the comparators were defined before it. Table R24 lists, for each method, the parameters, how and on which period they were chosen, and the trial-run values ($\bar{D}$ and the cost ratio to H1 at target 0.95, seeds 0 and 1).

VII. ADDITIONAL ANALYSES ADDED DURING REVISION

The analyses in this section were added during revision, after the confirmation results were known, with a plan fixed before any aggregate was computed; the main text discusses them in Section VII.

TABLE R16. Sensitivity to the observation radius and the horizon $T_{\max}$ on the held-out period. Hops, groups, and models are rebuilt for each setting. Columns as in Table S8, for evaluation seed 0; Seeds: number of the five evaluation seeds in which no group is below 0.94 in either segment; –: not seed-dependent (hindsight oracle). Split conformal calibration does not depend on the seed, so its entry repeats the result of seed 0.

	Segment A				Segment B				
Method	Recall	Worst	<0.94	Cost	Recall	Worst	<0.94	Cost	Seeds
Confirmation settings (radius 50 m, $T_{\max} = 600$ s): 176 sites; exit share 0.160									
hops A/B 264,918/243,238; structural miss A/B 0.009%/0.007%									
static	0.949	0.907	6	0.97	0.944	0.870	8	0.88	0/5
ACI	0.950	0.904	4	1.07	0.951	0.881	5	1.11	0/5
qr_ogd	0.950	0.943	0	1.20	0.950	0.921	1	1.18	2/5
mvp_log	0.950	0.942	0	1.18	0.950	0.927	0	1.17	3/5
hindsight oracle	0.950	0.909	6	1.00	0.950	0.876	5	1.00	–
Radius 30 m: 190 sites; exit share 0.182									
hops A/B 222,596/204,569; structural miss A/B 0.016%/0.019%									
static	0.949	0.897	6	0.97	0.944	0.865	7	0.87	0/5
ACI	0.950	0.905	6	1.04	0.950	0.882	6	1.05	0/5
qr_ogd	0.950	0.946	0	1.15	0.949	0.942	0	1.09	5/5
mvp_log	0.950	0.945	0	1.15	0.948	0.935	0	1.08	5/5
hindsight oracle	0.950	0.898	6	1.00	0.950	0.879	5	1.00	–
Radius 75 m: 170 sites; exit share 0.166									
hops A/B 253,130/233,040; structural miss A/B 0.006%/0.008%									
static	0.949	0.895	5	0.97	0.944	0.855	7	0.89	0/5
ACI	0.950	0.890	5	1.07	0.951	0.875	5	1.06	0/5
qr_ogd	0.950	0.929	0	1.19	0.950	0.915	1	1.15	3/5
mvp_log	0.949	0.924	1	1.14	0.950	0.926	0	1.14	4/5
hindsight oracle	0.950	0.895	5	1.00	0.950	0.861	5	1.00	–
$T_{\max} = 300$ s: 176 sites; exit share 0.170									
hops A/B 261,747/240,281; structural miss A/B 0.012%/0.014%									
static	0.948	0.911	6	0.97	0.944	0.903	7	0.90	0/5
ACI	0.949	0.913	5	1.03	0.950	0.916	5	1.03	0/5
qr_ogd	0.950	0.946	0	1.10	0.950	0.944	0	1.09	5/5
mvp_log	0.950	0.945	0	1.09	0.950	0.936	0	1.08	4/5
hindsight oracle	0.950	0.913	6	1.00	0.950	0.912	5	1.00	–
$T_{\max} = 900$ s: 176 sites; exit share 0.158									
hops A/B 265,567/243,779; structural miss A/B 0.008%/0.009%									
static	0.948	0.905	6	0.96	0.944	0.862	8	0.87	0/5
ACI	0.950	0.903	5	1.08	0.950	0.871	5	1.08	0/5
qr_ogd	0.950	0.928	0	1.22	0.950	0.910	1	1.19	4/5
mvp_log	0.950	0.918	1	1.18	0.948	0.898	1	1.14	0/5
hindsight oracle	0.950	0.905	5	1.00	0.950	0.864	5	1.00	–

TABLE R17. Ablation of the threshold scale on the exploratory periods (additional analysis). Cost relative to the hindsight oracle, averaged over the evaluation segments of each period; Mean: average of the two period means. Fail: segments in which some group is below 0.94. [a]With as many buckets as needed to keep the resolution of the bounded scale.

	Period 1		Period 2		
Scale	Cost	Fail	Cost	Fail	Mean
Uniform	1.80	0/2	2.05	0/3	1.92
Uniform + bound	1.47	0/2	1.83	0/3	1.65
Log + bound (mvp_log)	1.15	0/2	1.22	0/3	1.18
Log, no bound[a]	1.15	1/2	1.22	0/3	1.19
$a_{\min} = 0.001$	1.14	0/2	1.22	0/3	1.18
$a_{\min} = 0.005$	1.14	1/2	1.20	0/3	1.17

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TABLE R18. Sensitivity on the held-out period (evaluation seed 0; additional analysis). (a) Feedback fraction and delay, `mvp_log`. (b) Detection probability p_{det} with the original target (0.95) and the adjusted target $(1-\alpha)p_{\text{det}}$ (adj.); recall with respect to the true visibility set; cost relative to the hindsight oracle without detection errors.

(a) Feedback, mvp_log

	Worst group		<0.94		
Fraction, delay	A	B	A	B	Cost
10%, delayed	0.937	0.920	0	1	1.14
30%, delayed	0.942	0.927	0	0	1.18
100%, delayed	0.943	0.923	0	1	1.16
100%, immediate	0.936	0.930	0	0	1.17

(b) Detection errors

	Recall (true)		Worst group		
p_{det}, target	A	B	A	B	Cost
mvp_log					
0.95, 0.95	0.998	0.998	0.997	0.995	86.22
0.95, adj.	0.948	0.948	0.942	0.935	1.17
0.90, 0.95	1.000	1.000	0.999	0.999	136.98
0.90, adj.	0.944	0.944	0.937	0.931	1.17
static					
0.95, 0.95	0.998	0.997	0.994	0.996	27.78
0.95, adj.	0.947	0.942	0.903	0.869	0.89
0.90, 0.95	1.000	1.000	0.999	0.999	136.98
0.90, adj.	0.944	0.939	0.898	0.865	0.85

TABLE R19. Track custody on the held-out period (evaluation seed 0): share of chains whose first k hops are all hit, among the chains with at least k hops, by the track state of the first hop. Indep. 5: product of the per-hop hit probabilities of the first five hops, assuming independence.

		Segment A						Segment B					
Start	Method	$k=1$	2	3	4	5	Indep. 5	$k=1$	2	3	4	5	Indep. 5
All	chains	46,964	42,272	37,122	31,546	26,045		42,733	38,765	34,236	29,185	24,226	
	static	0.925	0.887	0.860	0.838	0.817	0.807	0.923	0.879	0.851	0.826	0.803	0.790
	ACI	0.928	0.891	0.865	0.844	0.822	0.812	0.933	0.895	0.869	0.846	0.824	0.814
	qr_ogd	0.948	0.907	0.874	0.845	0.819	0.810	0.952	0.908	0.874	0.844	0.817	0.809
	mvp_log	0.944	0.903	0.872	0.844	0.818	0.809	0.950	0.905	0.871	0.841	0.815	0.806
	hindsight oracle	0.927	0.890	0.864	0.843	0.822	0.812	0.932	0.893	0.867	0.844	0.822	0.812
Initiated	chains	45,910	41,419	36,423	30,972	25,590		41,887	38,077	33,650	28,716	23,854	
	static	0.925	0.887	0.861	0.839	0.818	0.809	0.924	0.881	0.852	0.827	0.805	0.792
	ACI	0.928	0.891	0.865	0.844	0.823	0.813	0.934	0.897	0.871	0.848	0.826	0.816
	qr_ogd	0.948	0.907	0.875	0.846	0.820	0.812	0.952	0.909	0.876	0.846	0.819	0.811
	mvp_log	0.944	0.904	0.873	0.845	0.819	0.810	0.951	0.906	0.872	0.842	0.816	0.807
	hindsight oracle	0.927	0.890	0.865	0.843	0.823	0.813	0.933	0.894	0.868	0.846	0.824	0.814
Re-acquired	chains	1,054	853	699	574	455		846	688	586	469	372	
	static	0.907	0.864	0.821	0.791	0.758	0.743	0.870	0.795	0.766	0.725	0.688	0.673
	ACI	0.904	0.864	0.828	0.796	0.758	0.738	0.881	0.805	0.778	0.746	0.710	0.693
	qr_ogd	0.943	0.880	0.833	0.787	0.749	0.745	0.921	0.842	0.804	0.753	0.726	0.710
	mvp_log	0.942	0.878	0.828	0.791	0.741	0.736	0.927	0.846	0.811	0.765	0.737	0.720
	hindsight oracle	0.909	0.866	0.825	0.798	0.769	0.752	0.876	0.802	0.778	0.742	0.707	0.689

TABLE R20. Camera outages on the held-out period, from 48 h after the outage start to the end of segment B (additional analysis). Random outages: mean over five draws of failed sites; Region: the 20 sites within a fixed radius of the busiest site. Recall: mean over all hops. <0.94: groups below the bar. Cost: relative to the hindsight oracle without outage. The outage-aware predictor was developed after the outage results with the original predictor were known; the exploratory periods give the same direction (Table R21).

		static			ACI			qr_ogd			mvp_log		
Outage	Pred.	Recall	<0.94	Cost	Recall	<0.94	Cost	Recall	<0.94	Cost	Recall	<0.94	Cost
Random 10%	base	0.862	25.0	0.80	0.949	5.4	34.19	0.934	12.0	9.90	0.932	14.4	14.87
	aware	0.918	22.4	0.90	0.950	4.0	1.87	0.949	2.4	2.05	0.945	2.4	2.30
Random 20%	base	0.820	25.0	0.72	0.950	4.2	45.49	0.922	21.6	17.07	0.923	19.2	19.80
	aware	0.898	24.6	0.87	0.950	4.0	2.83	0.949	2.0	2.86	0.944	3.6	3.30
Random 30%	base	0.757	25.0	0.66	0.950	4.6	67.47	0.903	23.6	29.67	0.885	22.8	21.49
	aware	0.873	24.8	0.87	0.950	2.8	4.02	0.948	1.8	3.85	0.944	4.6	4.22
Region (20 sites)	base	0.929	19	0.85	0.950	5	1.67	0.948	4	1.70	0.942	6	3.07
	aware	0.930	18	0.86	0.950	4	1.63	0.949	4	1.78	0.946	2	2.59

TABLE R21. Camera outages with the outage-aware predictor on the two exploratory periods (additional analysis), from 48 h after the outage start, which is the start of each period's second evaluation segment, to the end of its last evaluation segment; in exploratory period 1, this is the Christmas and New Year holiday segment. Columns as in Table R20. The full tables, including the original predictor and split conformal calibration, are Tables R6 and R7.

		ACI			qr_ogd			mvp_log		
Per.	Outage	Rec.	<.94	Cost	Rec.	<.94	Cost	Rec.	<.94	Cost
1	Rand. 10%	0.950	4.8	1.71	0.950	1.2	1.87	0.951	0.6	2.07
	Rand. 20%	0.950	5.0	2.92	0.950	1.6	3.17	0.948	1.4	3.52
	Rand. 30%	0.950	3.6	4.09	0.950	2.2	4.10	0.948	0.8	4.54
	Region	0.950	6	1.46	0.950	3	1.60	0.951	1	2.11
2	Rand. 10%	0.950	5.2	1.92	0.950	0.6	2.44	0.950	0.2	2.48
	Rand. 20%	0.950	4.2	2.92	0.950	0.8	3.68	0.949	0.2	3.81
	Rand. 30%	0.950	4.0	4.82	0.949	0.4	5.07	0.949	0.0	5.26
	Region	0.950	6	1.56	0.949	2	2.03	0.949	0	2.82

TABLE R22. Verdicts of the decision rules (Sec. 10.8 of the analysis plan) on the confirmation period and, applied after the confirmation, on the earlier held-out period (additional analysis; the held-out results were already known). S1: group against marginal calibration; S1 addition: online methods against `qr_groups`; S2: online methods against the refitted variants, one verdict per method, with the number of targets at which the online method was worse given per refitted variant; S4: rarest group, where "remains" means that the 95% lower bound of the re-acquisition shortfall was above zero at a majority of targets and "qualified" that it was not. Counts are over the five targets. The verdict words are those of the locked judgment file and the counts those of the file of estimates.

		Confirmation		Held-out	
Statement	Method	Judgement	Targets	Judgement	Targets
S1	`mvp_log`	kept	5/5	kept	5/5
S1	`qr_ogd`	kept	5/5	kept	5/5
S1 addition[a]	`mvp_log`	added	5, 0	added	5, 1
S1 addition[a]	`qr_ogd`	added	5, 0	not added	5, 5
S2	`mvp_log`	opposite		opposite	
worse than `qr_groups_refit`			5/5		4/5
worse than `qr_groups_refit7`			5/5		4/5
S2	`qr_ogd`	opposite		opposite	
worse than `qr_groups_refit`			3/5		3/5
worse than `qr_groups_refit7`			0/5		3/5
S4	`mvp_log`	remains	4/5	remains	5/5
S4	`qr_ogd`	qualified	0/5	remains	4/5

[a]Targets improved against Mondrian and against `qr_groups`.

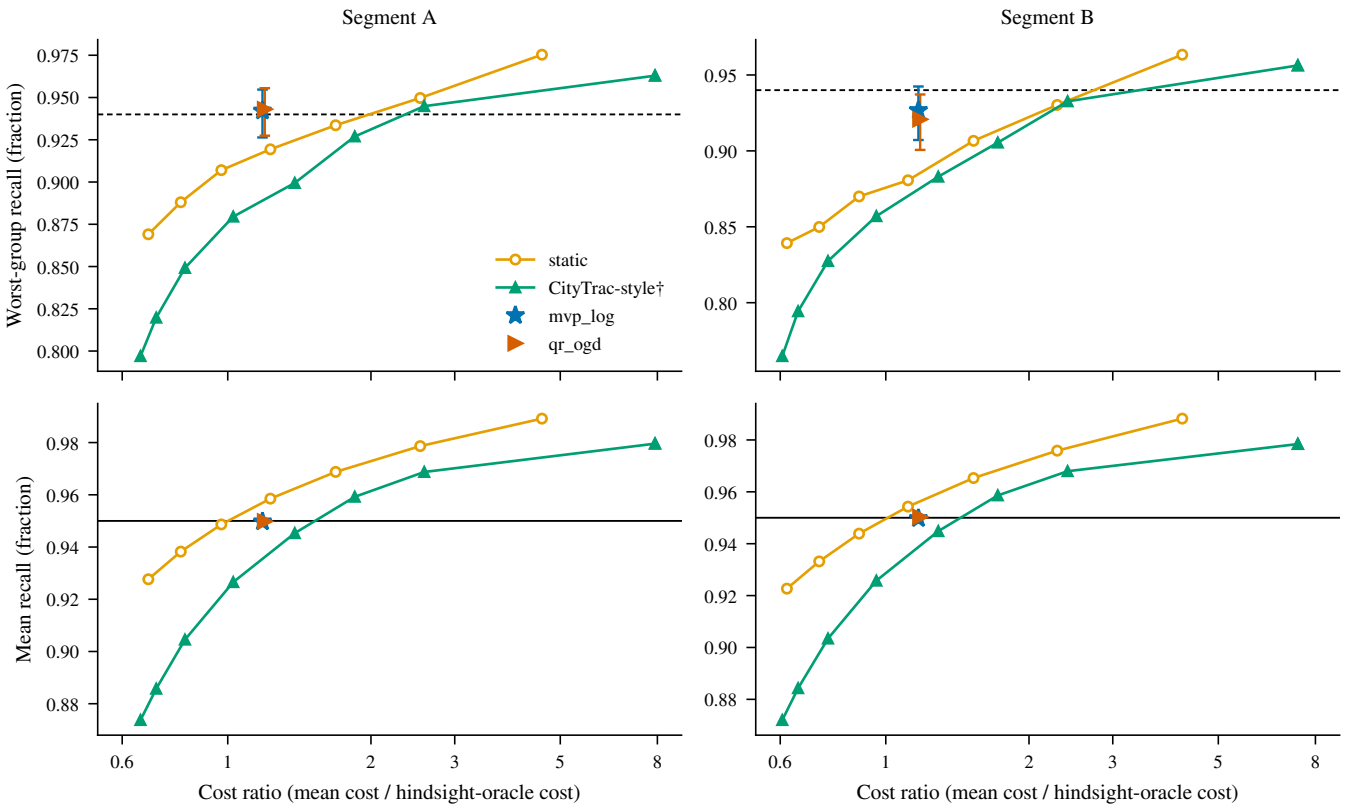


FIGURE R4. **Recall against cost on the held-out period (evaluation seed 0; additional analysis). Curves: `static` and the CityTrac-style rule with varying targets. Points: `mvp_log` and `qr_ogd`; vertical bars show the 95% Wilson interval of their worst group (re-acquisition hops); in segment A the bars overlap because the two costs nearly coincide. Top: worst-group recall, dashed line at the bar 0.94. Bottom: mean recall, line at the target 0.95. Logarithmic cost axis.**

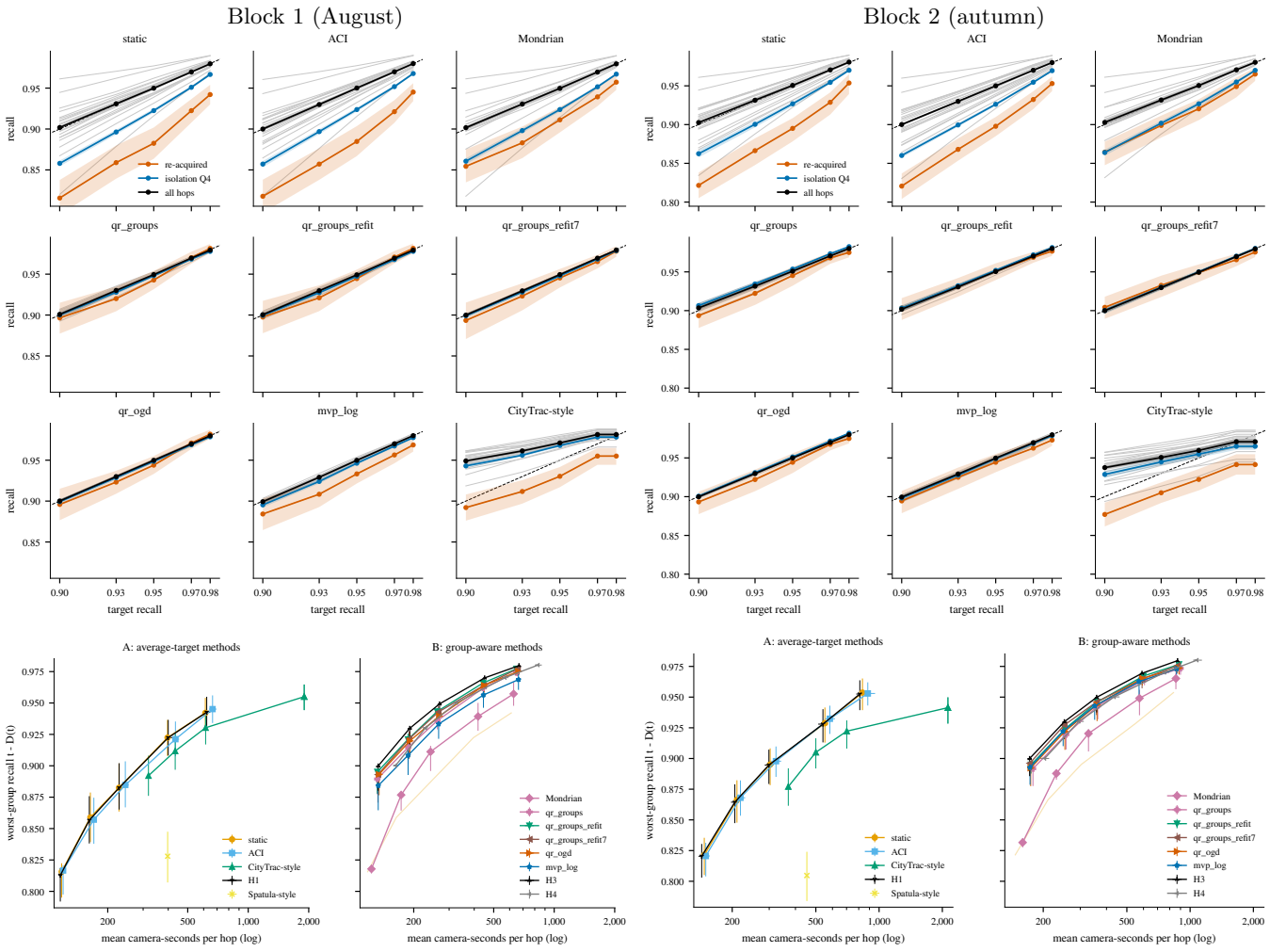


FIGURE R5. **Calibration (top) and cost (bottom) on each block of the confirmation period (left: 5 August to 1 September; right: 7 October to 3 November 2013), drawn as in Figs. 1 and 2.**

TABLE R23. Diagnostics on the confirmation period. Range over seeds of $\bar{D}$ and of the re-acquisition shortfall (pp) and its ratio to the interval width; three ACI variants ($\gamma = 0.0005$; $\gamma = 0.001$; group-wise ACI over time of day $\times$ first hop with $\gamma = 0.001$ and floor $\alpha/5$), means over seeds and blocks without intervals; share of hops at $a_{\min}$ for `mvp_log` at target 0.98; reachable upper bound of recall in the re-acquisition and isolation Q4 groups, both blocks pooled.

Method	Quantity	0.90	0.93	0.95	0.97	0.98
(a) Range over the 5 seeds, pp (ratio to the 95% interval width)						
`mvp_log`	$\bar{D}$	0.8–1.3 (0.32)	0.8–1.7 (0.52)	0.5–1.4 (0.67)	0.7–1.2 (0.42)	0.6–1.1 (0.40)
	re-acq. d	0.8–1.3 (0.23)	0.8–1.7 (0.42)	0.5–1.4 (0.54)	0.7–1.2 (0.40)	0.6–1.1 (0.38)
`qr_ogd`	$\bar{D}$	0.2–1.4 (0.79)	0.2–1.5 (0.78)	0.2–1.2 (0.80)	0.1–1.1 (1.92)	0.1–1.0 (1.49)
	re-acq. d	−0.3–1.4 (0.74)	−0.4–1.5 (0.91)	−0.1–1.2 (0.74)	−0.7–1.1 (1.69)	−0.5–1.0 (1.62)
`static`	$\bar{D}$	8.1–8.1 (0.00)	6.7–6.7 (0.00)	6.0–6.0 (0.00)	4.4–4.4 (0.00)	3.1–3.1 (0.00)
	re-acq. d	8.1–8.1 (0.00)	6.7–6.7 (0.00)	6.0–6.0 (0.00)	4.4–4.4 (0.00)	3.1–3.1 (0.00)
ACI	$\bar{D}$	7.8–8.2 (0.19)	6.4–6.8 (0.19)	5.6–5.9 (0.17)	4.2–4.3 (0.09)	2.9–3.1 (0.19)
	re-acq. d	7.8–8.2 (0.15)	6.4–6.8 (0.19)	5.6–5.9 (0.17)	4.2–4.3 (0.09)	2.9–3.1 (0.19)
Mondrian	$\bar{D}$	7.4–7.4 (0.00)	4.7–4.7 (0.00)	3.3–3.3 (0.00)	2.5–2.5 (0.00)	1.8–1.8 (0.00)
	re-acq. d	4.0–4.0 (0.00)	3.8–3.8 (0.00)	3.3–3.3 (0.00)	2.5–2.5 (0.00)	1.8–1.8 (0.00)
`qr_groups`	$\bar{D}$	0.4–1.3 (0.64)	0.4–1.7 (0.89)	0.3–1.2 (0.81)	0.2–1.2 (2.03)	0.2–1.0 (1.60)
	re-acq. d	−0.6–1.3 (0.82)	−0.5–1.7 (1.06)	−0.2–1.2 (0.81)	−0.7–1.2 (1.74)	−0.5–1.0 (1.66)
`qr_groups_refit`	$\bar{D}$	0.2–0.3 (0.09)	0.3–0.5 (0.22)	0.2–0.4 (0.21)	0.1–0.3 (0.41)	0.1–0.3 (0.46)
	re-acq. d	−0.4–0.1 (0.22)	0.0–0.5 (0.27)	−0.2–0.4 (0.37)	−0.2–0.3 (0.52)	0.0–0.3 (0.34)
`qr_groups_refit7`	$\bar{D}$	0.1–0.8 (0.61)	0.1–0.6 (0.40)	0.2–0.8 (0.60)	0.1–0.8 (0.83)	0.1–0.6 (0.72)
	re-acq. d	−0.6–0.8 (0.60)	−0.8–0.6 (0.67)	−0.9–0.8 (1.01)	−0.2–0.8 (0.91)	0.1–0.6 (0.56)
CityTrac-style	$\bar{D}$	1.7–1.7 (0.00)	2.2–2.2 (0.00)	2.4–2.4 (0.00)	2.3–2.3 (0.00)	3.3–3.3 (0.00)
	re-acq. d	1.7–1.7 (0.00)	2.2–2.2 (0.00)	2.4–2.4 (0.00)	2.3–2.3 (0.00)	3.3–3.3 (0.00)
Spatula-style	$\bar{D}$	8.6–8.6 (0.00)	11.6–11.6 (0.00)	13.6–13.6 (0.00)	15.6–15.6 (0.00)	16.6–16.6 (0.00)
	re-acq. d	8.6–8.6 (0.00)	11.6–11.6 (0.00)	13.6–13.6 (0.00)	15.6–15.6 (0.00)	16.6–16.6 (0.00)
(b) ACI variants (mean over seeds and the two blocks; no interval): recall / worst group / cost						
ACI $\gamma = 0.0005$ (main)	recall / worst / cost	0.900 / 0.819 / 135	0.930 / 0.863 / 195	0.950 / 0.891 / 284	0.970 / 0.927 / 509	0.980 / 0.949 / 774
ACI $\gamma = 0.001$	recall / worst / cost	0.900 / 0.818 / 136	0.930 / 0.863 / 197	0.950 / 0.893 / 289	0.970 / 0.927 / 516	0.980 / 0.949 / 800
ACI per hour $\times$ first hop	recall / worst / cost	0.900 / 0.822 / 135	0.930 / 0.880 / 196	0.950 / 0.914 / 283	0.970 / 0.944 / 496	0.980 / 0.962 / 767

(c) `mvp_log` hops at $a_{\min}$, target 0.98, seeds 0–4 (%): 0.0058, 0.0071, 0.0086, 0.0094, 0.0100
(d) Reachable upper bound of recall U_G: re-acquisition 0.9998, isolation Q4 0.9998

TABLE R24. Tuning history of each method (parameters, and how and on which period they were chosen) and the trial run on exploratory period 1 before the lock ($\bar{D}$ in pp and cost ratio to H1, target 0.95, seeds 0 and 1). The history before the pre-registration tag is taken from the research notes of that time and cannot be verified from the version history. After the trial run, only the recording code was corrected (`b5e10e6`); no method, comparator, or decision rule was changed.

				Trial run	
Method	Tuned parameters	Chosen on	Variants	$\bar{D}$ (pp)	Cost/H1
`static`	none	–	–	6.5	0.91
ACI	$\gamma \in \{0.005, 0.001, 0.0005\}$, floor $\alpha/5$, per hour $\times$ first hop; window 2,000 (initial value)	expl. 1 (virtual layout), then expl. 2 (virtual)	5, then 3 kept	5.5	1.07
Mondrian	cells (hour, first hop, hour $\times$ first hop; later speed and hour $\times$ first hop $\times$ speed, 40 cells); minimum 200	expl. 1, then expl. 2 (virtual)	5	3.9	0.98
`qr_groups`	log scale (linear scale gave negative τ); at most 80,000 rows	exploratory	2 scales	3.8	1.08
`qr_groups_refit`	as `qr_groups`, refitted daily, cumulative	defined before the trial run	1	0.9	1.10
`qr_groups_refit7`	as `qr_groups`, refitted daily, 7-day window	defined before the trial run	1	0.2	1.16
`qr_ogd`	$\gamma = 0.005$, clip 3	exploratory	1	2.6	1.16
`mvp_log`	log scale, $a_{\min} = 0.002$, $m = 40$	expl. 1 and 2 (real layout)	1	1.3	1.15
Spatula-style	share threshold 0.05, delay quantile 0.98	from the reading of the paper	–	16.1	1.48
CityTrac-style	delay window quantile 0.98, minimum 30 per key	from the reading of the paper	–	3.7	2.24
Predictor	hierarchy src $\rightarrow$ zone $\rightarrow$ hour $\rightarrow$ speed; $\lambda = 20$, $\varepsilon = 10^{-4}$, $\sigma = 1$ bin	expl. 1 and 2	5		
Groups, criteria	25 groups; bar 0.94 (first draft); cost bound 1.20	research records	–		
Trial run (pre-lock)	detection-time output compared in float64 like the score (the first trial run stopped on the consistency check); output code only, no method, comparator, or parameter changed (commit `b5e10e6`)	exploratory period 1	–		

Tuning history before the pre-registration tag (`prereg-v3`): research records (README versions 2–7 and internal deviation notes), not verifiable in the version history. Trial run: exploratory period 1, target 0.95, evaluation seeds 0–1, made before the lock of the confirmation plan.

TABLE R25. Feedback sensitivity on the confirmation period (target 0.95, both blocks pooled, mean over seeds): worst-group shortfall $\bar{D}$ and re-acquisition shortfall (pp), and cost relative to H2, with conservative 95% intervals. Label sets (i)–(iv) as in Section VII. The refit reads only labelled scores, so its rows for (iii) and (iv) equal those for (i) and (ii). A threshold above every predicted probability searches nothing and costs zero.

Label set	Method	$\bar{D}$	Re-acquisition d	Cost / H2
full	ACI	5.7 [4.7, 6.8]	5.7 [4.7, 6.8]	1.08 [1.07, 1.09]
	group-wise ACI	0.1 [−0.0, 0.8]	−0.1 [−0.9, 0.8]	1.10 [1.08, 1.11]
	mvp_log	1.0 [0.5, 1.9]	1.0 [0.2, 1.9]	1.18 [1.17, 1.19]
	qr_ogd	0.7 [0.2, 1.5]	0.6 [−0.2, 1.5]	1.20 [1.18, 1.21]
	refit	0.3 [0.2, 1.0]	0.2 [−0.6, 1.0]	1.18 [1.17, 1.20]
(i)	ACI	5.5 [4.5, 6.5]	5.5 [4.5, 6.5]	5.40 [5.10, 5.73]
	mvp_log	1.0 [0.5, 1.9]	1.0 [0.2, 1.9]	1.18 [1.17, 1.19]
	qr_ogd	0.7 [0.2, 1.5]	0.6 [−0.2, 1.5]	1.20 [1.18, 1.21]
	refit	2.4 [2.1, 3.3]	2.2 [1.2, 3.3]	1.23 [1.21, 1.24]
(ii)	ACI	5.7 [4.7, 6.7]	5.7 [4.7, 6.7]	1.23 [1.17, 1.31]
	mvp_log	1.1 [0.5, 2.0]	1.1 [0.3, 2.0]	1.18 [1.17, 1.19]
	qr_ogd	0.7 [0.2, 1.5]	0.6 [−0.2, 1.5]	1.20 [1.18, 1.21]
	refit	2.1 [1.8, 2.9]	1.9 [0.9, 2.9]	1.22 [1.21, 1.23]
(iii)	ACI	91.7 [88.6, 94.2]	90.6 [86.8, 93.4]	3.93 [1.41, 5.61]
	group-wise ACI	69.8 [63.8, 75.4]	3.0 [2.0, 4.1]	2.39 [1.78, 3.09]
	mvp_log	86.4 [81.5, 91.6]	21.4 [16.6, 26.6]	25.33 [17.61, 36.09]
	qr_ogd	93.0 [90.4, 94.8]	33.8 [30.5, 37.0]	16.95 [10.51, 21.13]
	refit[a]	2.4 [2.1, 3.3]	2.2 [1.2, 3.3]	1.23 [1.21, 1.24]
(iv)	ACI	40.5 [39.7, 41.3]	33.2 [31.3, 35.1]	1.03 [1.02, 1.05]
	group-wise ACI	20.8 [19.7, 21.8]	2.6 [1.6, 3.8]	1.10 [1.04, 1.17]
	mvp_log	22.3 [20.8, 23.7]	17.2 [15.0, 19.5]	1.27 [1.17, 1.38]
	qr_ogd	21.8 [21.5, 22.5]	6.2 [5.1, 7.4]	1.31 [1.23, 1.41]
	refit[a]	2.1 [1.8, 2.9]	1.9 [0.9, 2.9]	1.22 [1.21, 1.23]

[a](iii)/(iv) refit = (i)/(ii) refit by construction (the refit reads only labeled scores).

TABLE R26. Predictor sensitivity on the confirmation period (target 0.95, both blocks pooled, mean over seeds): worst-group shortfall $\bar{D}$ and re-acquisition shortfall (pp), and cost relative to the H2 of the same predictor, with conservative 95% intervals.

Predictor	Method	$\bar{D}$	Re-acquisition d	Cost / H2
src	static	6.0 [4.8, 7.2]	6.0 [4.8, 7.2]	1.00 [1.00, 1.00]
	ACI	6.1 [5.1, 7.2]	6.1 [5.1, 7.2]	1.05 [1.04, 1.06]
	Mondrian (40 cells)	3.2 [2.7, 4.2]	3.2 [2.3, 4.2]	1.02 [1.01, 1.02]
	qr_groups	0.6 [0.3, 1.1]	0.3 [−0.4, 1.0]	1.10 [1.08, 1.11]
	mvp_log	0.8 [0.4, 1.6]	0.8 [0.1, 1.6]	1.06 [1.05, 1.06]
	qr_ogd	0.6 [0.3, 1.1]	0.3 [−0.3, 1.1]	1.10 [1.09, 1.11]
src_zone	static	6.0 [5.0, 7.2]	6.0 [5.0, 7.2]	1.00 [1.00, 1.00]
	ACI	5.8 [4.8, 6.9]	5.8 [4.8, 6.9]	1.06 [1.05, 1.07]
	Mondrian (40 cells)	3.6 [2.8, 4.6]	3.6 [2.7, 4.6]	1.04 [1.03, 1.05]
	qr_groups	0.6 [0.3, 1.2]	0.4 [−0.3, 1.1]	1.12 [1.10, 1.13]
	mvp_log	1.3 [0.5, 2.3]	1.3 [0.4, 2.3]	1.10 [1.09, 1.11]
	qr_ogd	0.5 [0.2, 1.2]	0.5 [−0.2, 1.2]	1.13 [1.12, 1.14]
src_zone_hour	static	6.3 [5.2, 7.5]	6.3 [5.2, 7.5]	1.00 [1.00, 1.00]
	ACI	6.0 [5.0, 7.0]	6.0 [5.0, 7.0]	1.07 [1.06, 1.08]
	Mondrian (40 cells)	3.5 [2.8, 4.6]	3.5 [2.5, 4.6]	1.06 [1.06, 1.07]
	qr_groups	1.2 [0.6, 2.0]	1.0 [0.2, 2.0]	1.15 [1.14, 1.16]
	mvp_log	1.0 [0.5, 1.9]	1.0 [0.1, 1.9]	1.15 [1.14, 1.16]
	qr_ogd	1.0 [0.4, 1.9]	0.9 [0.2, 1.9]	1.17 [1.16, 1.18]
src_zone_speed	static	5.8 [4.7, 7.1]	5.8 [4.7, 7.1]	1.00 [1.00, 1.00]
	ACI	5.6 [4.5, 6.6]	5.6 [4.5, 6.6]	1.07 [1.06, 1.08]
	Mondrian (40 cells)	3.2 [2.9, 4.3]	3.2 [2.2, 4.3]	1.07 [1.06, 1.07]
	qr_groups	0.4 [0.2, 0.9]	0.1 [−0.7, 0.9]	1.17 [1.16, 1.18]
	mvp_log	1.2 [0.5, 2.1]	1.2 [0.3, 2.1]	1.15 [1.14, 1.16]
	qr_ogd	0.3 [0.1, 1.1]	0.2 [−0.6, 1.1]	1.18 [1.17, 1.19]
	qr_groups_refit	0.3 [0.2, 0.9]	0.1 [−0.7, 0.9]	1.16 [1.15, 1.17]
src_zone_hour_speed (primary)	static	6.0 [4.9, 7.2]	6.0 [4.9, 7.2]	1.00 [1.00, 1.00]
	ACI	5.7 [4.7, 6.8]	5.7 [4.7, 6.8]	1.08 [1.07, 1.09]
	Mondrian (40 cells)	3.3 [3.0, 4.4]	3.3 [2.4, 4.4]	1.08 [1.08, 1.09]
	qr_groups	0.8 [0.3, 1.5]	0.6 [−0.2, 1.5]	1.18 [1.17, 1.19]
	mvp_log	1.0 [0.5, 1.9]	1.0 [0.2, 1.9]	1.18 [1.17, 1.19]
	qr_ogd	0.7 [0.2, 1.5]	0.6 [−0.2, 1.5]	1.20 [1.18, 1.21]
	qr_groups_refit	0.3 [0.2, 1.0]	0.2 [−0.6, 1.0]	1.18 [1.17, 1.20]